

Final Report: Financial Q&A with Retrieval-Augmented Generation (RAG)

Abstract

Recent advances in large language models (LLMs) have enabled highly capable conversational systems. However, many of the best-performing models, such as GPT-4 and Gemini, are closed-source and opaque, limiting transparency and customizability. In this project, we explored an alternative pathway: using open retrieval-augmented generation (RAG) techniques and light prompt-based tuning to guide a commercial LLM (Gemini) to act more like a grounded rewriting agent rather than an unrestricted generator. Our goal was not to outperform proprietary LLMs, but to demonstrate that a RAG-driven approach, grounded in real financial Q&A data, can deliver human-like, helpful answers comparable in clarity and relevance, while maintaining openness and control. We built and deployed a finance-focused RAG chatbot using Streamlit, SentenceTransformer retrieval, Gemini 1.5 Flash for generation, and an LLM-based automatic evaluation system. Our findings show the strong potential of RAG architectures for creating transparent, accessible, and reproducible domain-specific assistants.

Introduction

Large Language Models (LLMs) like GPT-4, Gemini, and Claude have demonstrated remarkable capabilities in generating fluent and coherent answers across a wide variety of domains. However, despite their strengths, these leading models remain closed-source: their training data, fine-tuning processes, and architectural details are proprietary and inaccessible. This lack of transparency introduces significant challenges for practitioners seeking to build trustworthy, auditable, and domain-specific systems without relying entirely on opaque commercial platforms.

In this project, our goal was not to create a system that surpasses closed models like GPT-4 or Gemini. Rather, we set out to explore a different research question: Can we, using open retrieval methods, careful prompt tuning, and strategic control over input data, guide an LLM to act more like a reliable rewriting agent generating grounded, high-quality, clear financial answers comparable to those from closed systems?

A core focus of our design was minimizing hallucination, a known limitation of free-form LLM generation. By building a RAG pipeline where the builder maintains full control over the retrieval base, we ensured that generated answers were anchored in verified, curated financial knowledge. This human-in-the-loop control over the source data significantly enhanced the trustworthiness and predictability of the system, allowing us to maintain higher reliability standards than a typical free-generation LLM assistant.

Retrieval-Augmented Generation (RAG) thus emerged as the ideal architecture for this goal: it separates knowledge retrieval from language generation, enabling fine-grained control, transparency, and reproducibility. In sensitive domains like finance — where accuracy, simplicity, and user trust are essential — this approach provides a promising alternative to relying solely on closed, black-box LLMs.

Method

We built our Financial RAG system by combining open-source retrieval techniques with a lightweight prompt-tuned generation strategy.

Our question-answer pairs were sourced from the Sujet-Finance-Instruct-177k dataset ([Hugging Face link](#)), a large-scale corpus of finance-related instructional data. From this corpus, we extracted and curated approximately 2,000 QA pairs focused on personal finance, banking, investing, and economic principles. The data was cleaned to ensure concise, clear, and non-technical answers suitable for non-expert users.

Each question was embedded using the all-MiniLM-L6-v2 model from sentence-transformers, generating 384-dimensional dense vectors. Embeddings were precomputed and stored for fast cosine similarity retrieval during user queries. This approach allowed us to maintain full control over the data that the system retrieved from — ensuring transparency, reducing hallucination risks, and allowing us to anchor the chatbot’s behavior tightly to trusted content.

When a user submitted a question, we computed its embedding and retrieved the top-3 most similar QA pairs based on cosine similarity. These retrieved examples were passed into a carefully designed prompt template for Google’s Gemini 1.5 Flash model, with strict instructions:

- Use only the retrieved context.
- If insufficient information exists, explicitly say "I'm not sure based on the available data."
- Explain concepts simply and clearly, avoiding technical jargon.

This setup guided the LLM to behave as a rewriting agent — summarizing and rephrasing retrieved knowledge rather than inventing answers.

To monitor quality, we implemented automatic evaluation: after each answer, Gemini rated the response from 1 to 5 stars based on relevance, clarity, and helpfulness. This LLM-based scoring replaced subjective manual review and enabled consistent, scalable quality checks.

The complete system was deployed as an interactive web application using Streamlit, allowing users to ask questions, view live-streamed answers, see related retrieved Q&A pairs, and receive immediate feedback on answer quality.

Result

The final system performed reliably during informal testing, generating clear, concise, and relevant answers to a variety of finance-related user questions. While we did not conduct a formal evaluation over a large annotated test set, we used Gemini’s LLM-based scoring to rate the quality of each response after generation. This rating was based on a custom prompt asking Gemini to evaluate answers on relevance, clarity, and helpfulness, and return a 1 to 5 star score.

In most test cases, the system produced well-structured answers that scored 4 or 5 stars, confirming that the combination of retrieval grounding and prompt constraint led to meaningful outputs. The LLM also correctly deferred in situations where retrieved context lacked relevant information,

responding with: “I’m not sure based on the available data.” This behavior helped prevent hallucination, which was one of our key design goals.

Notably, our system did not require any training or fine-tuning. Both the embedding model (all-MiniLM-L6-v2) and generation model (Gemini 1.5 Flash) were used directly in inference mode. This reinforces that strong results can be achieved using off-the-shelf models, when combined with thoughtful data curation, retrieval quality, and prompt engineering.

Ethic Statement

This system is intended solely as an informational tool, not as a provider of professional financial advice. While it delivers finance-related answers in plain language, the generated content should not be interpreted as guidance for financial decision-making or investment strategy.

To promote transparency and trust, the model was designed to explicitly communicate uncertainty when sufficient context is not available — for example, responding with “I’m not sure based on the available data.” This behavior was reinforced through prompt instructions and retrieval-based grounding.

We collected no personal or user-specific data during interaction, and all model responses are generated in real-time based on public, pre-curated financial information. The QA dataset used, although cleaned and filtered for clarity, may reflect regional (e.g., U.S.-centric) or general knowledge biases present in the original corpus. These limitations were considered acceptable for the scope of this educational and exploratory system.

Conclusion

This project demonstrates that it is possible to build a reliable, explainable, and user-friendly financial Q&A system using open retrieval pipelines and prompt-controlled LLMs — without any model training or proprietary fine-tuning.

Rather than trying to outperform state-of-the-art closed-source LLMs, our objective was to explore how a builder-controlled RAG architecture could deliver comparable quality while offering greater transparency, reduced hallucination, and user trust. By constraining the language model to act as a rewriting agent over retrieved content, we achieved grounded responses that remained faithful to curated source material.

Key takeaways include:

- Prompt design and retrieval control can significantly reduce hallucination.
- Off-the-shelf components (no training needed) are sufficient for building practical domain-specific assistants.
- Transparency in the form of showing retrieved context improves user confidence.
- Automatic evaluation via LLM scoring offers scalable and consistent quality monitoring.

This approach is especially valuable for academic, educational, or small-scale production use cases where full model training is not feasible, but trust and clarity are essential.

Future directions may include expanding the retrieval base, comparing open LLMs (like Mixtral or Mistral) in place of Gemini, and evaluating performance across more diverse financial user profiles.

AI Usage Declaration and Sharing

This project used the following pre-trained models and APIs:

- Embedding Model: sentence-transformers/all-MiniLM-L6-v2 (Hugging Face)
- QA Dataset: Subset of [Sujet-Finance-Instruct-177k](#)
- Generative Model: Google's Gemini 1.5 Flash via the google-generativeai Python SDK
- Evaluation Model: Same Gemini model used with a custom scoring prompt
- Deployment: Streamlit app hosted on Streamlit Community Cloud

We did not train or fine-tune any models. All components were used in inference mode. API keys were stored securely using Streamlit Secrets, and the system is fully reproducible using openly available libraries and data.

The source code includes modules for:

- Embedding QA data
- Retrieving similar QAs
- Prompting Gemini for generation and scoring
- Streamlit-based user interface

The full pipeline is transparent and modular, designed to be replicable for educational or research purposes.